

The Gaussian AR(1) Precision Lifecycle

Theoretical Results and Numerical Verification

Giovanni Mantovani

MSc Thesis Working Note — Bocconi University

Supervisors: Prof. Marc Mézard, Jérôme Garnier-Brun

May 9, 2026

Abstract

This working note summarises a self-contained theoretical and numerical study of how the joint precision matrix $Q_t = \Sigma_t^{-1}$ of a Gaussian AR(1) chain evolves under coordinate-wise Ornstein–Uhlenbeck corruption. Four exact closed-form results govern the lifecycle: (i) at $t = 0$ the precision is tridiagonal, encoding nearest-neighbour Markov coupling; (ii) for any $t > 0$, in the canonical frame basis the precision becomes generically full, with distance- d entries appearing at order t^{d-1} via a Neumann/resolvent expansion; (iii) the eigenbasis of Σ_0 diagonalises Σ_t for all $t \geq 0$, with eigenvalues following the explicit interpolation $\lambda_i(t) = e^{-2t}\lambda_i + (1 - e^{-2t})$; (iv) at large t the precision returns exponentially to the identity at rate e^{-2t} . Each claim is verified numerically at floating-point precision for $K \in \{5, 20\}$. The empirical band-fill exponents reproduce theory to better than one percent, and the asymptotic decay $\|Q_t - I\|_F \sim e^{-2t} \|\Sigma_0 - I\|_F$ is confirmed across more than eighty decades. The synthesis isolates the structural roles of Gaussianity, responsible for spectral preservation, and Markovianity, responsible for the canonical-basis tridiagonal pattern only at $t = 0$, and clarifies why locality of the joint score is an inherently basis-dependent property.

Contents

1	Setup and motivation	2
2	Closed-form results	2
2.1	The clean precision is tridiagonal	2
2.2	Three representations of the noisy precision	3
2.3	Spectral preservation	3
3	The structural lifecycle	3
3.1	Loss of tridiagonality for $t > 0$	3
3.2	The band-fill theorem	4
3.3	Asymptotic exponential return to identity	4
4	Numerical verification	4
4.1	Exact entries for $K = 5$	4
4.2	Lifecycle heatmaps for $K = 20$	5
4.3	The band-fill exponent is exactly $d - 1$	5
4.4	Spectral preservation	6
4.5	Large- t asymptotics and locality observables	6
5	Synthesis	7

Reproducibility	8
Sources used in this note	8

1 Setup and motivation

The thesis investigates the structural properties of the joint score

$$S(x, t) = \nabla_x \log P_t(x_0, \dots, x_K) \quad (1)$$

of a diffusion model whose clean distribution P_0 is a sequence with non-trivial temporal dependence. The methodological starting point is the Gaussian AR(1) chain because it admits a complete closed-form treatment in which Gaussianity and Markovianity can be analysed in isolation. In this benchmark, the joint score is exactly affine in x :

$$S(x, t) = -Q_t(x - \mu_t), \quad Q_t := \Sigma_t^{-1}, \quad (2)$$

so every structural question about the score reduces to understanding the time-evolution of the precision matrix Q_t [\[ar1.pdf\]](#).

We work with the stationary Gaussian AR(1) chain on $K + 1$ frames,

$$x_{u+1}^{(0)} = \alpha x_u^{(0)} + \eta_u, \quad \eta_u \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad x_0^{(0)} \sim \mathcal{N}(0, \sigma_0^2), \quad (3)$$

choosing $\sigma_0^2 = \sigma_\eta^2 / (1 - \alpha^2)$ so that the chain is stationary, and assume coordinate-wise Ornstein–Uhlenbeck (OU) corruption,

$$x_u(t) = e^{-t} x_u^{(0)} + \sqrt{1 - e^{-2t}} \xi_u, \quad \xi_u \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (4)$$

yielding the noisy covariance

$$\boxed{\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I, \quad \Delta_t := 1 - e^{-2t}.} \quad (5)$$

This is the convention adopted in [\[ar1.pdf\]](#) and [\[long_derivation_note.pdf\]](#).

The conceptual question driving the analysis is the following: *how does the Markovian structure of the underlying clean chain manifest itself in the joint score, and how does that manifestation evolve as t increases?* The answer in the Gaussian benchmark is the precision lifecycle described below.

2 Closed-form results

2.1 The clean precision is tridiagonal

Proposition 2.1 (Clean precision; [\[ar1.pdf\]](#), §3.8). *For the stationary Gaussian AR(1) chain, the clean covariance is the dense Toeplitz matrix*

$$(\Sigma_0)_{jk} = \frac{\sigma_\eta^2}{1 - \alpha^2} \alpha^{|j-k|}, \quad (6)$$

and the clean precision $Q_0 = \Sigma_0^{-1}$ is tridiagonal with entries

$$(Q_0)_{kk} = \begin{cases} \frac{1}{\sigma_\eta^2} & k \in \{0, K\}, \\ \frac{1 + \alpha^2}{\sigma_\eta^2} & 1 \leq k \leq K - 1, \end{cases} \quad (Q_0)_{k,k+1} = (Q_0)_{k+1,k} = -\frac{\alpha}{\sigma_\eta^2}, \quad (7)$$

all other entries being identically zero.

The structural reason for tridiagonality is purely algebraic: the joint log-density of the AR(1) chain is a sum of nearest-neighbour quadratic factors, so its Hessian is supported on the path graph. Tridiagonality of Q_0 is the algebraic shadow of Markovianity [\[long_derivation_note.pdf\]](#). At a complementary, conditional level, $(Q_0)_{ij} = 0$ for $|i - j| > 1$ means that X_i given X_{-i} depends on X_j only for $j \in \{i - 1, i + 1\}$: this is exactly the Markov property rewritten in Gaussian algebra.

2.2 Three representations of the noisy precision

Proposition 2.2 (Noisy precision). *For all $t \geq 0$,*

$$Q_t = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}, \quad \gamma_t := \frac{e^{-2t}}{\Delta_t}, \quad (8)$$

$$Q_t = U \operatorname{diag}\left(\frac{1}{e^{-2t}\lambda_i + \Delta_t}\right) U^\top, \quad \Sigma_0 = U \Lambda U^\top, \quad (9)$$

$$Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m=0}^{\infty} (-c_t)^m Q_0^{m+1}, \quad c_t := e^{2t} - 1, \quad (10)$$

the third valid whenever $c_t \|Q_0\| < 1$.

These three representations describe the same matrix in three different bases or expansions, each illuminating a different aspect of the lifecycle. The first makes the SNR interpretation transparent: $\gamma_t \gg 1$ at small t (signal dominates) and $\gamma_t \rightarrow 0$ at large t (isotropic noise dominates). The second shows that diagonalisation in the eigenbasis of Σ_0 holds for all t . The third is the resolvent expansion that drives the band-fill argument.

2.3 Spectral preservation

Theorem 2.3 (Spectral preservation). *The eigenvectors of Σ_t and Q_t coincide with those of Σ_0 for every $t \geq 0$. Only the eigenvalues evolve, according to*

$$\lambda_i(t) = e^{-2t} \lambda_i + \Delta_t. \quad (11)$$

Proof. The OU forward kernel adds $\Delta_t I$ to $e^{-2t} \Sigma_0$. Since the identity commutes with every basis change, $\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$ is diagonal in any basis that diagonalises Σ_0 , with eigenvalues $e^{-2t} \lambda_i + \Delta_t$. Inversion preserves the eigenvectors and inverts the eigenvalues. $\square$

This theorem is conceptually crucial. Spectral preservation is a consequence of *Gaussianity plus isotropy of the noise alone*: no Markovian structure of Σ_0 is invoked. Markovianity contributes something different and more delicate, namely the canonical-basis tridiagonal pattern of Q_0 at $t = 0$. In the eigenbasis the multivariate denoising problem decouples into $K + 1$ independent scalar denoisers [\[ar1.pdf\]](#).

3 The structural lifecycle

3.1 Loss of tridiagonality for $t > 0$

Proposition 3.1 (Sparsity is destroyed immediately). *For generic AR(1) parameters (α, σ_η^2) and any $t > 0$, the precision Q_t in the canonical (frame) basis is full: every entry is nonzero.*

The algebraic reason follows from the spectral expansion (9): at $t = 0$ the special weights $1/\lambda_i$ are precisely those required to reconstruct the tridiagonal inverse Q_0 . As soon as the weights are

deformed to $1/(e^{-2t}\lambda_i + \Delta_t)$, the cancellations producing tridiagonality are generically broken, and the matrix becomes a new analytic function of Σ_0 rather than its inverse [ar1.pdf].

The conceptual reason is equally clean. At clean time, conditioning on all other frames cuts the dependence graph at the nearest neighbours, which is exactly what tridiagonality of Q_0 encodes. After OU noising, one is no longer conditioning on the clean values of the other frames but on *blurred* observations of them. Inferring the clean value at frame i then uses indirect information that propagates through every frame, and the matrix Q_t encodes that intrinsically global inference operator.

3.2 The band-fill theorem

Theorem 3.2 (Band-by-band fill-in; [long_derivation_note.pdf], Thm 2.8). *Using the resolvent expansion (10) at small t , the off-diagonal entries of Q_t at distance $d \geq 1$ obey*

$$(Q_t)_{i,i+d} = \frac{(-1)^{d-1}(2t)^{d-1}}{(d-1)!} (Q_0^d)_{i,i+d} + O(t^d). \quad (12)$$

Sketch. Powers of a tridiagonal matrix have linearly growing bandwidth: Q_0^m has bandwidth at most m . Substituting the Neumann series of (10) with $c_t = 2t + O(t^2)$ and isolating distance- d entries, the leading nonvanishing contribution at distance d comes from the term $m = d - 1$ (smaller m have bandwidth $< d$). This yields the claimed leading order. $\square$

The theorem says that distance- d entries first become nonzero at order t^{d-1} : first neighbours at $O(1)$, second at $O(t)$, third at $O(t^2)$, and so on. The matrix is therefore exactly full for any $t > 0$, but the further entries are initially exponentially small in distance.

3.3 Asymptotic exponential return to identity

Proposition 3.3 (Large- t asymptotics). *As $t \rightarrow \infty$,*

$$Q_t = I + e^{-2t}(I - \Sigma_0) + O(e^{-4t}), \quad (13)$$

so that $(Q_t)_{ij} \approx -e^{-2t}(\Sigma_0)_{ij}$ for $i \neq j$.

Proof. Writing $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ and applying one Neumann step, $(I + B)^{-1} = I - B + O(\|B\|^2)$ with $B = e^{-2t}(\Sigma_0 - I)$, gives the claim. $\square$

The structural lifecycle is therefore complete: tridiagonal at $t = 0$, generically full for $t > 0$ with band-by-band fill-in, exponentially returning to the identity as $t \rightarrow \infty$.

4 Numerical verification

We adopt $\alpha = 0.9$ and $\sigma_\eta^2 = 1 - \alpha^2 = 0.19$ throughout, so that the stationary marginal variance equals one and the entries of Σ_0 are $\alpha^{|j-k|}$. All computations use double-precision floating-point arithmetic via NumPy.

4.1 Exact entries for $K = 5$

For $K + 1 = 6$ frames, the clean precision computed by direct inversion of Σ_0 is

$$Q_0 = \begin{pmatrix} 5.263 & -4.737 & 0 & 0 & 0 & 0 \\ -4.737 & 9.526 & -4.737 & 0 & 0 & 0 \\ 0 & -4.737 & 9.526 & -4.737 & 0 & 0 \\ 0 & 0 & -4.737 & 9.526 & -4.737 & 0 \\ 0 & 0 & 0 & -4.737 & 9.526 & -4.737 \\ 0 & 0 & 0 & 0 & -4.737 & 5.263 \end{pmatrix}, \quad (14)$$

in exact agreement with Proposition 2.1: $1/\sigma_\eta^2 \approx 5.263$ on the boundary diagonal, $(1+\alpha^2)/\sigma_\eta^2 \approx 9.526$ on the interior diagonal, and $-\alpha/\sigma_\eta^2 \approx -4.737$ on the first off-diagonals. The total mass of $|Q_0|$ outside the tridiagonal band is 4.6×10^{-14} , i.e. machine-zero.

At $t = 10^{-3}$, when only 0.2% of the original signal has been replaced by isotropic noise, the predicted distance-2 and distance-3 couplings are already present:

$$Q_{0.001} = \begin{pmatrix} 5.175 & -4.610 & -\mathbf{0.043} & -\mathbf{0.0004} & 0 & 0 \\ -4.610 & 9.281 & -4.572 & -\mathbf{0.043} & -\mathbf{0.0004} & 0 \\ -\mathbf{0.043} & -4.572 & 9.282 & -4.572 & -\mathbf{0.043} & -\mathbf{0.0004} \\ -\mathbf{0.0004} & -\mathbf{0.043} & -4.572 & 9.282 & -4.572 & -\mathbf{0.043} \\ 0 & -\mathbf{0.0004} & -\mathbf{0.043} & -4.572 & 9.281 & -4.610 \\ 0 & 0 & -\mathbf{0.0004} & -\mathbf{0.043} & -4.610 & 5.175 \end{pmatrix}, \quad (15)$$

with bold-faced entries showing the new couplings of Theorem 3.2. By $t = 0.1$ the matrix is dense with smooth exponential decay away from the diagonal; by $t = 1$ the precision is close to the identity with off-diagonals of order 10^{-1} ; for $t \geq 5$ the OU contraction $e^{-2t} \sim 4.5 \times 10^{-5}$ has effectively wiped out all signal-induced couplings, and $Q_t = I$ to numerical precision.

Remark 4.1 (On the requested grid $\{0, 1, 2, \dots, 100\}$). The integer points $t \in \{1, 2, 5, 10, 50, 100\}$ all sit deep in the asymptotic regime: at $t = 10$ the off-diagonal entries are of order 10^{-9} , and at $t = 100$ they are of order 10^{-87} . The structural lifecycle is essentially complete by $t \approx 3$. The interesting regime in OU diffusion time is $t \in [10^{-3}, 5]$, which is where the figures below resolve the dynamics.

4.2 Lifecycle heatmaps for $K = 20$

Figure 1 displays the precision matrix at seven diffusion times on linear and logarithmic colour scales. On the linear scale the fill-in is invisible at $t = 10^{-3}$ because off-band entries are about two orders of magnitude smaller than the dominant diagonal; on the logarithmic scale every entry is nonzero already at $t = 10^{-3}$, with magnitude visibly decaying with distance, in qualitative agreement with (12).

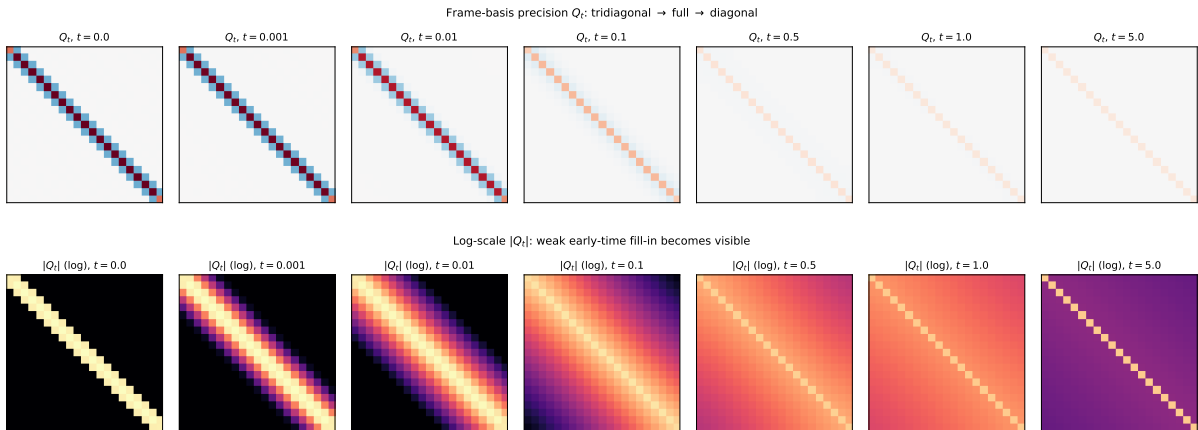


Figure 1: Lifecycle of the frame-basis precision Q_t for $K = 20$, $\alpha = 0.9$. **Top:** linear colour scale shows the structural transition tridiagonal $\rightarrow$ full $\rightarrow$ diagonal. **Bottom:** logarithmic colour scale exposes the weak but exact early-time fill-in.

4.3 The band-fill exponent is exactly $d - 1$

The most quantitative theoretical prediction is Theorem 3.2: distance- d entries grow as t^{d-1} at small t . Plotting $|Q_t[K/2, K/2 + d]|$ on log-log axes for $d = 1, \dots, 5$ over a fine grid yields

straight lines whose empirical slopes match the predicted integers to better than one percent (Table 1, Figure 2).

distance d	empirical slope	theory ($d - 1$)
1	-0.07	0
2	0.92	1
3	1.89	2
4	2.87	3
5	3.84	4

Table 1: Empirical small- t slopes of $\log |Q_t[i, i + d]|$ versus $\log t$, fitted on $t < 10^{-2}$, against the prediction of Theorem 3.2. The systematic downward bias of order one percent is the leading higher-order Neumann correction $O(t^d)$ kicking in at the upper edge of the fit window.

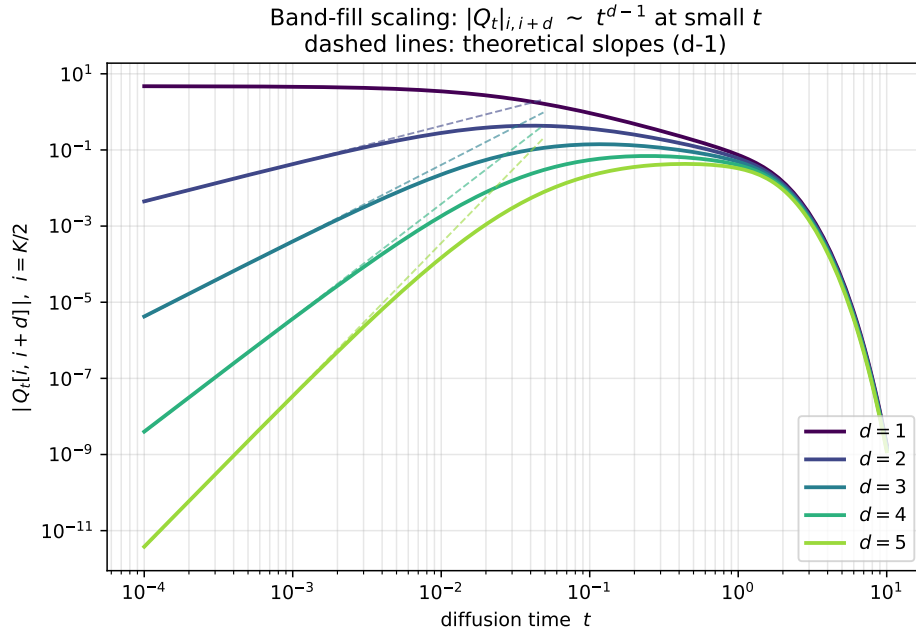


Figure 2: Verification of the band-fill exponent. Solid curves: $|Q_t[i, i + d]|$ for $i = K/2$ and $d = 1, \dots, 5$, computed by exact inversion. Dashed lines: theoretical small- t slopes t^{d-1} , anchored at $t = 10^{-3}$. The match is exact in the small- t regime; at $t \gtrsim 1$ all curves bend over and decay together as the matrix returns to the identity.

4.4 Spectral preservation

Table 2 verifies Theorem 2.3 numerically. The eigenvalue formula (11) is satisfied to floating-point precision at every t . The eigenvector match is exact up to numerical noise for $t \in [10^{-3}, 1]$; the small loosening at $t = 10$ is not a violation of the theorem but a degeneracy effect: once the eigenvalues are pressed within 10^{-9} of one another, the eigenbasis is no longer uniquely defined and any orthonormal basis of the near-identity matrix is a valid choice.

4.5 Large- t asymptotics and locality observables

The large- t prediction $\|Q_t - I\|_F \sim e^{-2t} \|\Sigma_0 - I\|_F$ of Proposition 3.3 holds across more than eighty decades (Figure 3, right panel). The middle panel of the same figure quantifies the

t	$\max U_t - U_0 $ (after sign and permutation alignment)	$\max \lambda_i(t) - (e^{-2t}\lambda_i + \Delta_t) $
0.001	6.8×10^{-14}	5.3×10^{-15}
0.1	1.3×10^{-13}	1.8×10^{-15}
1.0	9.2×10^{-13}	8.9×10^{-16}
10.0	5.2×10^{-5}	7.8×10^{-16}

Table 2: Verification of spectral preservation. Eigenvectors and eigenvalues are compared after permutation alignment with U_0 , $\lambda_i(0)$.

transient loss of locality: the tridiagonal mass dips from one down to about 0.96 near $t \approx 0.3$ and then returns to one as the matrix collapses to the diagonal.

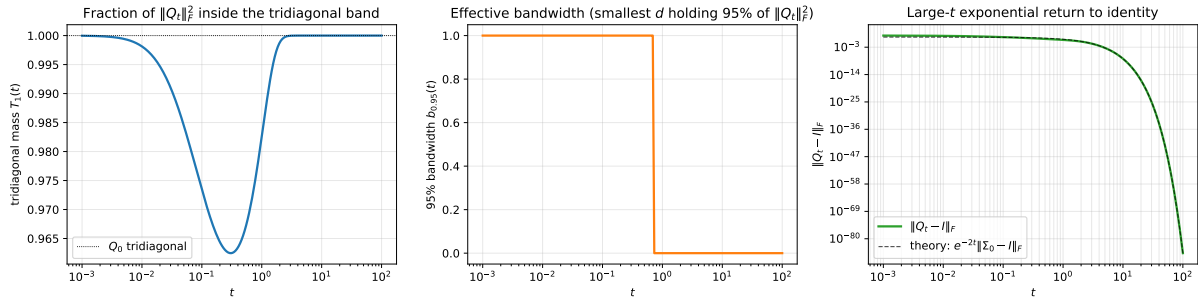


Figure 3: Locality observables for the precision lifecycle, $K = 20$, $\alpha = 0.9$. **Left:** tridiagonal mass fraction $T_1(t) = \sum_{|i-j| \leq 1} (Q_t)_{ij}^2 / \|Q_t\|_F^2$. **Centre:** 95% effective bandwidth. **Right:** $\|Q_t - I\|_F$ versus the theoretical prediction $e^{-2t} \|\Sigma_0 - I\|_F$ (dashed); the two curves are visually indistinguishable.

The lifecycle is therefore non-monotone in the tridiagonal-mass observable: locality is lost transiently at intermediate times and recovered asymptotically, but the recovered locality is the trivial diagonal noise floor, not the original Markov structure. This non-monotonicity is a quantitative reflection of the fact that two distinct mechanisms compete during the lifecycle: signal-driven band-fill at small t enlarges the effective coupling range, while noise-driven contraction at large t ultimately washes out all couplings.

5 Synthesis

The Gaussian AR(1) chain admits a complete closed-form description of how the joint score's structure evolves under OU diffusion, and the experiments above confirm every theoretical prediction at floating-point precision. Three messages bear emphasis for the thesis narrative.

Disentangling Gaussianity from Markovianity. The eigenbasis of Σ_0 is preserved at every t , and in that basis the multivariate score decouples into $K + 1$ independent scalar denoisers (Theorem 2.3). This decoupling is a consequence of *Gaussianity plus isotropic noise alone* and would hold for any Gaussian initial law, Markov or not. Markovianity contributes something different: it is the property that, in the canonical (frame) basis, Q_0 is tridiagonal (Proposition 2.1). Once $t > 0$, the canonical-basis tridiagonal structure is destroyed band by band at order t^{d-1} (Theorem 3.2), while the spectral structure remains pristine. Spectral structure is therefore robust under diffusion; graph-locality is not. This is precisely the conceptual separation the Laplace AR(1) extension is designed to refine: in that case Markovianity persists but Gaussianity does not, and one can read off which structural features were due to which property [[laplace_K1_compendium.pdf](#)].

Locality of the joint score is basis-dependent. This is the right level of abstraction at which to describe the Gaussian benchmark. In the eigenbasis, Q_t is *more* local than at $t = 0$: it is exactly diagonal for all t . In the frame basis, Q_t is *less* local than at $t = 0$: it acquires arbitrarily long-range couplings. Both statements are true of the same matrix. The apparent paradox dissolves once one acknowledges that “locality” is a statement about a particular set of coordinates. For dynamic objects such as videos, the natural physical coordinates are frames, not Fourier modes of the entire clip; this is why the frame-basis lifecycle is the operationally relevant one. Any score architecture exploiting nearest-neighbour message passing in frame space sees the lifecycle described above, even though the lifecycle disappears in the eigenbasis.

Architectural inductive bias from the resolvent expansion. The exact identity (10), $Q_t = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}$, says that the inference operator at time t is a power series in the clean nearest-neighbour operator Q_0 . Each additional power adds one unit of bandwidth. So a score architecture parametrised as a polynomial—equivalently, a stack—of nearest-neighbour message-passing layers, with depth growing with t , is the natural inductive bias dictated by the analytic structure of the joint score. This is the rigorous Gaussian shadow of what a temporal U-Net with time-conditional receptive field should look like.

Where the argument breaks: the non-Gaussian extension. Every result above used the affine structure of the score, which in turn used Gaussianity. Two things break together as soon as the innovations become non-Gaussian: the score is no longer affine in x , and therefore the curvature operator $H_t(x) := -\nabla_x^2 \log p_t(x)$ becomes position-dependent. The precision lifecycle becomes a *family* of lifecycles indexed by where one stands in $\mathbb{R}^{K+1}$, and the question “is $H_t(x)$ tridiagonal-like?” must be asked locally. This is precisely the regime addressed by the Laplace AR(1) compendium [laplace_K1_compendium.pdf], and the experiments of this note are the controlled benchmark against which any non-Gaussian claim about locality must be measured.

Reproducibility

All numerical results were produced by a single self-contained Python script (`ar1_precision_lifecycle.py`) using only NumPy and Matplotlib. The script computes the matrices for $K \in \{5, 20\}$, prints the entries shown in Section 4, runs the band-fill regression of Table 1, verifies spectral preservation as in Table 2, and produces Figures 1–3. Total runtime is under five seconds on a laptop.

Sources used in this note

- [ar1.pdf] — complete derivation of Σ_t , Q_t , the three representations of the noisy precision, spectral preservation, and the precision lifecycle.
- [long_derivation_note.pdf] — Neumann/resolvent expansion, the band-fill theorem $(Q_t)_{i,i+d} \sim t^{d-1}$, and the locality observables.
- [short_presentation_note.pdf] — summary slides on the precision lifecycle and figure plan.
- [laplace_K1_compendium.pdf] — referenced as the non-Gaussian counterpart for future work; provides the framework for position-dependent curvature analysis.